Exploring Best Practices for GenAI and LA Dashboards

Introduction & Theoretical Foundations

- **Welcome & Introductions:** Brief introduction to the team and the workshop objectives
- **The State of Learning Analytics Dashboards:**
  - Brief introduction to the current landscape of LADs and trends (Verbert et al. 2020, Sahin & Ifenthaler 2021)
- **GenAI for Learning Analytics:**
  - Main opportunities and challenges (Susnjak 2022)
  - Contextualising GenAI in the LA cycle (Yan et al. 2023)
- **Overview of GenAI:**
  - Live poll, from the basics to more complex concepts of GenAI
  - Fundamentals of GenAI:
    - **ChatGPT/Gemini**: AI chatbots powered by large language models (LLMs) developed by OpenAI (ChatGPT) and Google DeepMind (Gemini), capable of generating human-like text responses
    - **LLM (Large Language Model)**: A deep learning model trained on vast amounts of text data to understand and generate natural language, used for tasks like text generation, summarization, and code completion
    - **Training and Inference**: *Training*: The process of teaching an LLM by feeding it massive datasets and adjusting weights to learn language patterns; *Inference*: The phase where the trained model generates responses based on new input
    - **Fine-tuning**: The process of training a pre-trained LLM on a smaller, specialized dataset to improve its performance on domain-specific tasks
    - **Prompt**: A user-inputted query or instruction that guides an LLM's response
    - **System/Developer Prompt**: A hidden or predefined instruction that sets the behaviour, constraints, or personality of an LLM before user interaction
    - **Temperature**: A parameter controlling randomness in LLM responses—higher values (e.g., 1.0) increase diversity, while lower values (e.g., 0.2) make responses more deterministic
    - **RLHF (Reinforcement Learning from Human Feedback) & DeepSeek-R1 Zero RL**: *RLHF*: A training technique where human evaluators rank model outputs to refine responses based on user preferences; *DeepSeek-R1 Zero RL*: A self-improving reinforcement learning method for LLMs, reducing reliance on human-labeled data
    - **RAG (Retrieval-Augmented Generation) and Similar Techniques**: A method where an LLM retrieves external knowledge from a database or documents to generate more accurate and up-to-date responses
    - **Chain-of-Thought (CoT) and Similar Techniques**: A reasoning approach where an LLM generates intermediate steps before the final answer, improving problem-solving and complex reasoning. Variants include Self-Consistency and Tree-of-Thought
    - **Multimodal/Vision Models**: AI models that process multiple data types (e.g., text, images, video, audio), enabling tasks like image captioning, object detection, and text generation based on visuals

Best Practices

A first attempt (Joshi 2024, Kew et al. 2024, Trevisan 2024) :

1. Clarity and Simplicity: Ensure the interface is uncluttered, focusing on essential metrics. Use AI to automatically highlight critical data points, reducing cognitive load.
2. User-Centric Design: Engage end-users in the design process to understand their needs. Use AI-driven surveys or feedback tools to tailor the interface to user preferences.
3. Effective Visualization: Utilize AI algorithms to generate intuitive visualizations like heatmaps or trend lines. This helps users quickly grasp data trends without confusion.
4. Customizable Views: Implement AI-powered customization options that adapt based on individual teaching strategies, offering dynamic and relevant insights.
5. Accessibility Features: Integrate AI-driven features like voice commands and screen readers to ensure the dashboard is usable by all, including individuals with disabilities.
6. Data Privacy and Security: Use AI for real-time monitoring of data security, ensuring compliance with regulations through automated checks.
7. Collaborative Partnerships: Foster partnerships between educators, technologists, and policymakers. Leverage AI tools to facilitate collaborative problem-solving and decision-making.
8. Reflective Analytics: AI can analyze historical data to provide insights on teaching effectiveness, supporting reflective practice and suggesting improvements.
9. Seamless Integration: Use AI to integrate with other educational tools, enabling comprehensive analysis from various data sources without manual intervention.
10. Continuous Testing and Iteration: Apply AI-driven A/B testing to refine the dashboard based on user feedback, ensuring it evolves to meet changing needs.

Example Dashboards:

- **ELAT - edX Log Analysis Tool** (Valle Torre et al. 2020)
  - Description of the type of data displayed (e.g., student engagement metrics, assessment results, learning resource usage)
  - Proof-of-Concept Integration: Screenshots and context in conversation, graph analysis and course design implications
- **TeamSlides and VizChat** (Echeverria et al. 2024, Yan et al. 2024)
  - VizChat introduction and how it addresses common challenges like cognitive overload, trust issues, and data visualization literacy
  - VizChat's key features:
    - Contextualized, AI-generated explanations for visualizations
    - Use of vision with GPT4v to process screenshots of dashboards
    - RAG + documentation for reliable information
    - Main capabilities: clarification, personalized responses, multi-visualization integration, and data collection explanations
  - VizChat hints at a paradigm shift from exploratory to explanatory LADs.

Participant Project Sharing

- **Project Sharing:** Participants briefly shared a LAD or LA project they are working on where they envision using GenAI.
- **Group Formation:** Using the authors of the systems as team leads, we formed working groups.

Group Review and Brainstorming

- **Dashboard Assessment**
  - **Scenario**: Target users, data source, pedagogical foundation, maturity level (idea, prototype, pilot, small/large scale implementation)
  - **Characteristics**: Where is the dashboard used and what are the main components? What's a typical workflow?
  - **Adoption**: What are the major barriers: technical infrastructure, management buy-in, user buy-in, trust? Who are the key facilitators?
  - **Challenges**: What are the initial challenges in the dashboard use?
- **GenAI Integration:** How can GenAI be integrated into the project? What specific functionalities could it provide?
  - **Potential Benefits:** What are the expected benefits of using GenAI in your context (e.g., improved user understanding, personalized insights, enhanced engagement)?
  - **Challenges & Risks:** What potential challenges or ethical concerns might arise from using GenAI in your project? How could these be mitigated?
  - **Improvement Opportunities:** Are there ways to refine your GenAI implementation based on the discussion and concepts covered so far?
  - **Technical Challenges**: What would be the main barriers from a technical perspective? Is it training/literacy (human side) or model capabilities (AI side)?

Interactive Prototype & Prompt Engineering

- **Open WebUI Playground**
  - **Prompt Engineering Activity:** crafting effective prompts to elicit desired responses from GenAI
  - Example prompts: Consider the purpose and the different stakeholders (students, teachers, counsellors)
    - Student - "Help me to interpret this chart?"
    - Student - "Based on this visualization, what can I do to improve my grade in the mid-term exam?"
    - Teacher - "How was this data collected?"
    - Student - "What does this chart tell me about my learning progress compared to my peers?" (if applicable)
    - Administrator - "What are the limitations of this data?"
  - Discuss the quality of the responses
- **Discussion:**
  - Group presentation: Explain the GenAI integration
    - How were the opportunities for GenAI in the dashboard?
    - How well did the GenAI responses address the owner's challenges?

- What are the strengths and limitations of using GenAI in this system?
- How could the interaction be improved?

- Discussion on the importance of user-centred design when developing GenAI-powered LADs, as well as the capabilities of GenAI for rapid prototyping and testing
- What about pitfalls, such as hallucinations or responsibility/accountability?

Wrap-up & Next Steps

- **Key Takeaways:** Summary of the main points of the workshop (the potential of GenAI for LADs, the importance of careful design and ethical considerations, and the value of user-centred approaches)
  - During both the project sharing and the interactive session, we collect comments and aggregate them here. The main concerns, challenges, opportunities and interesting opinions will be aggregated in the dashboard.
- **Call to Action:** Explore GenAI further and consider how you can apply these concepts to your own work. Keep in touch, especially to tell us more about your LA and GenAI integration for future Npuls communication and research.
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